Siddhesh Bhadale

Backend & Platform Engineer | Distributed Systems, Networking & Real-Time Infrastructure
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SUMMARY

Backend and platform engineer with 2 years shipping production systems for port, customs and enterprise customers. Technical lead for a remote-inspection platform — Node.js and Go services behind Kong, MongoDB data model, Docker Swarm to Kubernetes migration, React and React Native clients — now running at multiple port sites. Also authored the Rust/libp2p transport that reaches those on-premises services with no open port, public IP or VPN.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Rust
Production exposure: Kotlin, Swift
Backend: Node.js, NestJS, Express.js, REST API design, WebSockets, Kong API Gateway, Nginx, microservices, modular monolith, authentication & RBAC, multi-tenancy
Clients: React Native, React, Tauri
Data & Infrastructure: MongoDB (schema design, indexing, aggregation pipelines), MySQL, Redis, Docker, Docker Swarm, Kubernetes, Linux, reverse proxy & TLS, Make and shell release automation, on-premises deployment
Systems & Networking: Rust, libp2p, Tokio, QUIC/TCP, NAT traversal (DCUtR, circuit relay), Noise, Yamux, UniFFI, WebRTC/WHIP, ONVIF/PTZ

EXPERIENCE

Junior Mobile Application Developer Dec 2024 — Jun 2026
EaseMyAI *Thane, India*

- Technical lead for TeamStream, a remote-inspection platform for port and customs operations: owned discovery, architecture and tech-stack selection, then shipped the backend, React web app and React Native mobile apps with a four-engineer team — reviewing pull requests, planning sprints and coordinating stakeholders. Also delivered enterprise work across aviation, ports and logistics — two React Native apps and an Electron desktop app.
- Designed the backend as independently deployable Node.js and Go services behind Kong and Nginx — authentication, RBAC, session scheduling, notifications, recording, media-session orchestration — with the MongoDB schema, indexes and aggregation pipelines.
- Owned deployment end to end: containerised every service, ran it on Docker Swarm, then migrated to Kubernetes; Make and shell automation collapsed a multi-step release into one command.
- Supported UAT involving approximately **200 participants across 5 live-inspection sessions** with multiple live camera feeds each, on one Mac mini M4 (24 GB); deployed at multiple port sites, and showcased at the **India AI Impact Summit 2026**, demonstrated there to the Union Minister for Electronics & IT.

Mobile Application Developer — Intern Aug 2024 — Dec 2024
Applied Cloud Computing *Mumbai, India*

- Built production features for a top-10 Indian AMC's investor and distributor mutual-fund apps in React Native, integrating 34+ REST APIs for folio creation, SIP purchase and modification and portfolio reporting, plus Razorpay payment flows.

Full Stack Developer — Freelance Dec 2023 — Jul 2024
Independent *Remote*

- Delivered web and mobile products for 20+ clients as sole engineer across design, backend, deployment and handover; the largest serves 1,000 daily active users.

SELECTED PROJECTS

Peer-to-Peer Connectivity Layer for On-Premises Systems *EaseMyAI | Sole engineer*
Rust, libp2p 0.56 (DCUtR, circuit relay, Noise, QUIC/TCP), Tokio, UniFFI, Node.js, Express, MongoDB, Docker, systemd

- Port, airport, defence and government customers run our inspection software strictly on-premises and will not expose a public IP, open a port or run a VPN. Built a libp2p transport in Rust that dials the on-prem server directly — about 19,600 lines of Rust, TypeScript, Kotlin and Swift, written solo.
- Reserved a circuit-relay path for guaranteed reachability, then upgraded to a direct connection via DCUtR hole punching where the NAT pair allowed it — degrading to relayed delivery on symmetric NAT and CGNAT rather than failing.
- Exposed the tunnel as a loopback proxy on `127.0.0.1` so REST, WebSocket, chat, file transfer and live video ran over P2P with no application rewrite — including a hand-written Rust HTTP/1.1 proxy covering hop-by-hop headers, chunked encoding, CORS, reverse-proxy and CONNECT modes.
- Built the Node.js/MongoDB bootstrap service (registration, heartbeat, session tokens, QR pairing) plus an operator dashboard for hole-punch rate, relay circuits and RTT, and shipped one Rust core to 4 Android ABIs and 3 iOS targets via UniFFI. Tested P2P connectivity with **40+ peer sessions** against one on-prem server under a controlled test workload; now backs two products.

OPEN SOURCE

- **jiangdongguo/AndroidUSBCamera #753** — upstream PR to a 2.7k-star UVC camera engine, unmaintained since Sep 2024, that gated our body-camera feature. Upgraded Gradle 6.7.1 to 8.13 and NDK 27 / SDK 36 / Kotlin 2.0.21; dropped the prebuilt `.so` blobs and rebuilt `libuvc`, `libnative` and `libausbc` from source via `externalNativeBuild`, enabling the 16 KB memory page size now required by Play Store. **+316 / −197 lines across 63 files in one commit**, addressing 4 open upstream issues (#735, #744, #745, #746). **Referenced downstream** by nolte/kamerplanter-android #19 (merged Aug 2026), which cites #753 as the upstream fix for the same alignment failure.

EDUCATION

B.E. Computer Engineering — Savitribai Phule Pune University (SPPU), Pune 2020 — 2024
Languages: English, Hindi, Marathi